Level	Member	Research responsibility	Number of studies
Level 2	Rai	Résumé information extraction and NLP	2
Level 2	Fiona	Skill and job requirement normalization	2
Level 2	Tine	Applicant-facing and reciprocal job recommendations	2

Required output for each member

Each member should complete **two separate RRL analysis entries** using the same documentation format.

Required column	Information to extract
1. Author(s) and Year	Complete APA 7 reference
2. Research Objective	Problem investigated
3. Algorithm / Model	Methods proposed, implemented, or reviewed
4. Dataset	Source, size, and characteristics
5. Evaluation Metrics	Performance measurements used
6. Key Findings	Actual research results
7. Limitations / Research Gap	Identified weaknesses or limitations
8. Relevance to AYUDA	How the study supports our system
9. Implementation Decision	Adopt, adapt, benchmark, or exclude, with justification

Important: Members should read the actual papers rather than relying solely on abstracts or AI-generated summaries. For systematic reviews, document the methods reviewed rather than claiming the authors implemented all those algorithms.

Rai: Résumé information extraction and NLP

RRL 7 — **Automated Resume Parsing: A Natural Language Processing Approach**

IEEE CSITSS · 2023

This paper investigates résumé parsing through a combination of Named Entity Recognition (NER), keyword matching, and regular expressions to identify details such as names, contact information, education, skills, and experience.

IEEE Xplore

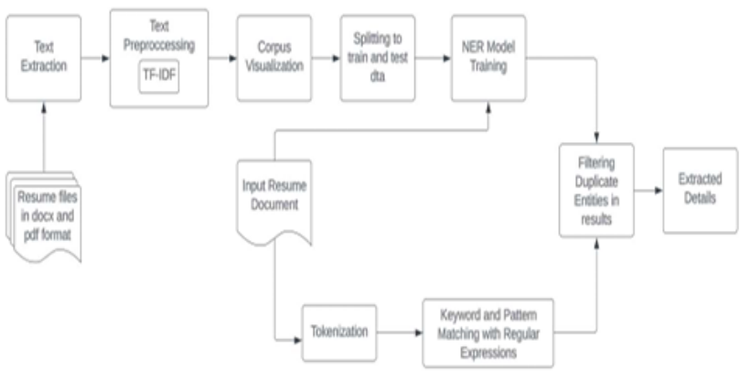
What Rai should investigate: How NER and rule-based extraction complement one another, which fields are extracted, how accuracy is evaluated, and whether the approach can handle varied résumé formats.

AYUDA relevance: Supports converting submitted résumés into structured candidate profiles before job matching.

Read RRL 7

Automated Resume Parsing: A Natural Language Processing Approach

Author(s) and Year	Sougandh et al. (2023) Sougandh, T. G., K, S. S., Reddy, N. S., & Belwal, M. (2023). Automated resume parsing: a natural language processing approach. International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), 1–6.

	https://doi.org/10.1109/csitss60515.2023.10334236
Research Objective	The study aims to develop an automated résumé parsing system that integrates Named Entity Recognition (NER) and Regex-based keyword and pattern matching to extract and organize structured candidate information, including personal details, skills, education, and work experience, from different résumé formats.
Algorithm / Model	 **Fig 3: Flow chart of the Resume Parser for extracting Details.** The study implements a hybrid résumé parsing approach combining a SpaCy-based Named Entity Recognition (NER) model with keyword and pattern matching using Regular Expressions (Regex). The NER model is trained on annotated résumé data to identify candidate-related entities, while Regex extracts information using predefined patterns. The extracted outputs are combined, filtered for duplicate or irrelevant values, and presented as structured candidate information. Supporting tools include PDFMiner, docx2txt, Python, Pandas, and YAML.
Dataset	The researchers assessed the NER model using a testing dataset and demonstrated the extraction of candidate information through sample outputs and a graphical interface. However, no numerical evaluation metrics, such as precision, recall, F1-score, or accuracy, were reported for the proposed parser.
Evaluation Metrics	The researchers assessed the NER model using a testing dataset and demonstrated the extraction of candidate information through sample outputs and a graphical interface. However, no numerical evaluation metrics, such as precision, recall, F1-score, or accuracy, were reported for the proposed parser.
Key Findings	The proposed résumé parser demonstrated the extraction of candidate information, including names, contact details, skills, education, work experience, languages, and CGPA, using a combination of NER and Regex-based pattern matching. The extracted information was presented in a structured format, with a graphical interface supporting résumé comparison. Although the authors described the approach as reliable and efficient, the study did not report numerical accuracy or processing-time measurements to substantiate these claims.
Limitations / Research Gap	Although the study presents an automated résumé parser combining NER and Regex-based extraction, its evaluation lacks detailed dataset information, including the number of résumés and training/testing split. Furthermore, the researchers did not report numerical performance metrics, making it difficult to assess the parser's extraction accuracy, efficiency, and consistency across different résumé formats.
Relevance to AYUDA	The study is relevant to AYUDA's proposed résumé processing and candidate profiling module because it demonstrates how NER and Regex-based extraction can organize candidate information from résumé documents into structured records. This approach could reduce repetitive manual encoding, support efficient retrieval and reuse of previously sourced candidates, and provide verified candidate information for the proposed candidate-to-job matching system.
Implement Decision	**Adapt / Further Investigate**. The study's combination of NER and

	Regex-based extraction may serve as a foundation for AYUDA's proposed résumé processing module. However, additional capabilities for information standardization, recruiter verification, and structured candidate profiling would need to be investigated to support candidate rediscovery and explainable job matching. The use of pretrained NLP models and rule-based extraction will be considered as a potentially feasible approach, subject to further research and validation.
Synthesis	Sougandh et al. (2023) demonstrate the use of NER and Regex-based extraction to convert résumé information into structured candidate data. While the approach provides a potential foundation for automated candidate profiling, the study does not establish quantitative extraction performance or demonstrate how the extracted information can support explainable candidate-to-job matching. For AYUDA, the approach may be adapted as a supporting technology, with further research needed on data standardization, verification, and integration with the proposed DSS.

RRL 8 — Resume Parsing Across Multiple Job Domains Using a BERT-Based NER Model

IEEE AICS · 2023 conference, indexed 2024

The researchers fine-tuned a BERT-based NER model using annotated résumés from IT, HR, consultancy, and engineering. The paper emphasizes the importance of dataset size, annotation quality, and domain variation.

IEEE Xplore

What Rai should investigate: BERT fine-tuning, annotated training data, entity extraction metrics, generalization across job domains, and limitations of small datasets.

AYUDA relevance: Helps assess whether a trained NLP model is practical compared with a simpler extraction pipeline.

Read RRL 8

Resume Parsing Across Multiple Job Domains Using a BERT-Based NER Model

Author(s) and Year	Srivastava and Greaney (2023) Srivastava, M., & Greaney, P. (2023). Resume parsing across multiple job domains using a BERT-Based NER model. Irish Conference on Artificial Intelligence and Cognitive Science (AICS), 1–4. https://doi.org/10.1109/aics60730.2023.10470917
Research Objective	The study aims to develop and evaluate a fine-tuned BERT-based Named Entity Recognition (NER) model for extracting structured candidate information from résumés across multiple job domains using limited annotated training data, while investigating how dataset size and annotation quality influence model performance and generalization.

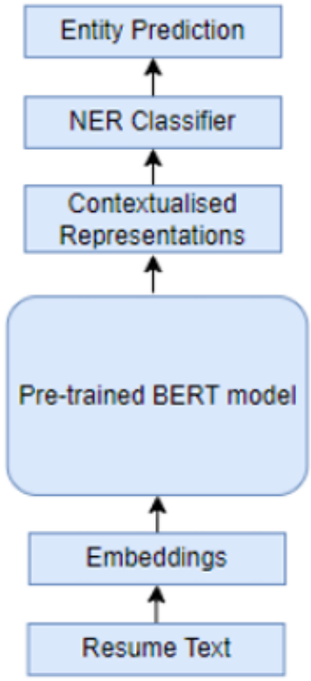
<table>
<tr>
<td>Algorithm / Model</td>
<td>


Fig. 1. Fine-tuned BERT model for resume NER

The study implements a fine-tuned BERT-based Named Entity Recognition (NER) model using SpaCy's training pipeline. The pretrained BERT model generates contextual representations of résumé text, which are processed by an NER classifier consisting of a feedforward network and softmax activation to identify four entity categories: job title, experience, skill, and education. The model is fine-tuned using manually annotated résumé samples from multiple job domains.
</td>
</tr>
<tr>
<td>Dataset</td>
<td>
TABLE I
RESULTS OF TUNED MODELS
<table>
<tr><th>No. of Samples</th><th>No. of Entities</th><th>Metric</th><th>Train Set</th><th>Validation Set</th></tr>
<tr><td rowspan="3">100</td><td rowspan="3">4940</td><td>F1 Score</td><td>0.9285</td><td>0.7002</td></tr>
<tr><td>Precision</td><td>0.9220</td><td>0.7167</td></tr>
<tr><td>Recall</td><td>0.9351</td><td>0.6844</td></tr>
<tr><td rowspan="3">140</td><td rowspan="3">6216</td><td>F1 Score</td><td>0.9364</td><td>0.6994</td></tr>
<tr><td>Precision</td><td>0.9292</td><td>0.6678</td></tr>
<tr><td>Recall</td><td>0.9436</td><td>0.7342</td></tr>
<tr><td rowspan="3">180</td><td rowspan="3">7425</td><td>F1 Score</td><td>0.9316</td><td>0.7714</td></tr>
<tr><td>Precision</td><td>0.9155</td><td>0.7539</td></tr>
<tr><td>Recall</td><td>0.9482</td><td>0.7896</td></tr>
</table>

The study utilized approximately 2,400 résumé samples sourced from LiveCareer.com, originally in PDF format. From this collection, 180 résumés across four job domains—Human Resources, Information Technology, Consultancy, and Engineering—were manually annotated for job titles, skills, education, and experience. Three experiments were conducted using 100, 140, and 180 annotated résumés, containing 4,940, 6,216, and 7,425 entities, respectively. Each experiment used an 80:20 training-validation split.
</td>
</tr>
<tr>
<td>Evaluation Metrics</td>
<td>The study evaluated the fine-tuned BERT-based NER model using precision, recall, and F1-score across three experiments involving 100, 140, and 180 annotated résumés. The 180-sample experiment achieved the highest overall validation F1-score of 0.7714, with a precision of 0.7539 and recall of 0.7896. Entity-level evaluation showed the highest validation F1-score for job titles (0.9474) and the lowest for education (0.7101).</td>
</tr>
<tr>
<td>Key Findings</td>
<td>The study demonstrated that fine-tuning a pretrained BERT-based NER model can support résumé information extraction across multiple job domains using</td>
</tr>
</table>

	limited annotated training data. Increasing the dataset to 180 résumés improved the overall validation F1-score to 0.7714, although overfitting remained evident. Entity-level results showed varying performance, with job titles achieving the highest validation F1-score (0.9474) and education the lowest (0.7101). The findings emphasize the importance of dataset size, annotation quality, and model optimization in résumé information extraction.
Limitations / Research Gap	The study was limited by a relatively small annotated dataset of 180 résumés covering four job domains, resulting in persistent overfitting and varying extraction performance across entity categories. The researchers also emphasized the importance of annotation quality and consistency, noting that larger, more diverse datasets and improved annotation procedures are needed to strengthen model generalization. Increasing the dataset size alone may not fully resolve the observed overfitting.
Relevance to AYUDA	The study is relevant to AYUDA's proposed résumé processing and candidate profiling module because it demonstrates the potential of BERT-based NER to extract structured candidate information, including job titles, skills, education, and work experience, across multiple job domains. Its findings emphasize the importance of extraction accuracy and data quality, which are essential for developing reliable candidate profiles that can support candidate rediscovery and the proposed explainable job-matching DSS. The study also provides a basis for evaluating the feasibility of pretrained NLP approaches with recruiter verification rather than immediately adopting custom model fine-tuning.
Implement Decision	**Compare / Adapt**. The study's BERT-based NER approach will be considered as a research alternative rather than directly adopted. Given the challenges of limited annotated training data and overfitting, AYUDA's proposed system will initially investigate using an existing pretrained NLP model supplemented by rule-based extraction and manual verification of candidate information. This approach may reduce the need for custom model training while supporting structured candidate profiling and the proposed matching DSS. Its suitability will require further evaluation and client validation.
Synthesis	Sougandh et al. (2023) and Srivastava and Greaney (2023) demonstrate the potential of NLP-based approaches for extracting structured candidate information from unstructured résumés. Sougandh et al. combined Named Entity Recognition (NER) with Regex-based pattern matching to extract multiple candidate fields, while Srivastava and Greaney employed a fine-tuned BERT-based NER model to recognize job titles, skills, education, and experience across different job domains. Although the latter provided measurable extraction performance, its findings revealed challenges involving limited annotated data, overfitting, and varying accuracy across entity categories. For AYUDA, these studies support investigating an adaptable résumé-processing approach that combines pretrained NLP with rule-based extraction and recruiter verification. However, further evaluation is needed to establish whether this approach can reliably produce structured candidate profiles suitable for candidate rediscovery and explainable job matching.

Fiona: Skill and job requirement normalization

RRL 9 — Joint Extraction and Classification of Danish Competences for Job Matching

Li & Lioma · ECIR / Springer · 2023

This study proposes a BERT-like model that both extracts and classifies skills, occupations, and knowledge from job postings. It uses structured competence labels and reports improved efficiency compared with the models evaluated in the study.

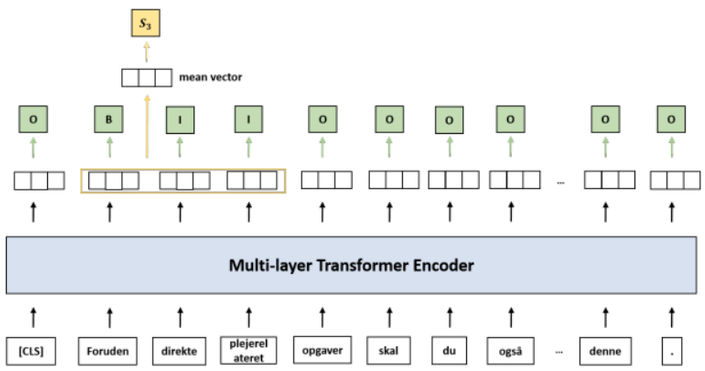
University of Copenhagen Research Portal

What Fiona should investigate: Competence extraction, classification, ESCO labels, the relationship between extraction and normalization, and whether the method transfers to English-language recruitment records.

AYUDA relevance: Supports converting different descriptions of skills and qualifications into comparable categories.

Read RRL 9

Joint Extraction and Classification of Danish Competences for Job Matching

Author(s) and Year	Li, Q., & Lioma, C. (2023). Joint extraction and classification of Danish competences for job matching. In J.Kamps, L. Goeuriot, F. Crestani, M. Maistro, H. Joho, B. Davis, C. Gurrin, U. Kruschwitz, & A. Caputo (Eds.), *Advances in information retrieval: 45th European Conference on Information Retrieval, ECIR 2023* (Lecture Notes in Computer Science, Vol. 13981, pp. 475–483). Springer. https://doi.org/10.1007/978-3-031-28238-6_38
Research Objectives	The study aimed to develop a BERT-based model that automatically extracts and classifies Danish competences, including skills, occupations, and knowledge areas, from job postings using ESCO labels. The model was designed to support job matching and improve recruiter efficiency by processing competence information accurately and quickly.
Algorithm/Model	 **Fig. 2.** The architecture of the joint ESCO extraction and classification model. The study proposed a single BERT-based model for jointly extracting and classifying Danish competences. It used DaBERT and DaJobBERT as text encoders. The model first used Named Entity Recognition to detect competence terms and then classified them using ESCO categories, such as skills, occupations, and knowledge. The model was trained jointly but performed extraction and classification sequentially during prediction. It was compared with SkillSpan plus Kompetencer, a separate detection-and-classification model, and an all-class NER model.

Data Set	The study used 217,661 Danish sentences from the Jobindex database, including job-posting descriptions and candidate education and work-experience information. The data were annotated with ESCO labels for skills, occupations, knowledge, and other competences. The dataset was divided into training, validation, and test sets using an 8:1:1 ratio. The data were provided by Jobindex and were not publicly available.
Evaluation Metrics	The study evaluated **ESCO competence extraction** using the **F1-score**, which measured whether tokens were correctly identified as part of an ESCO competence. It evaluated **ESCO competence classification** using the **weighted Macro-F1 score**, which measured whether extracted competences were assigned to the correct ESCO categories. The study also measured efficiency using the **average processing time per job**, based on 100 randomly selected job postings. Model size and inference time were additionally reported for comparison, but model size itself is not an evaluation metric.
Key Findings	The proposed joint BERT-based model achieved the strongest competence classification performance among the compared approaches and processed job postings more than 50% faster than the SkillSpan + Kompetencer pipeline. Joint training improved classification performance, while competence extraction remained comparable to other models. The results also showed that DaBERT and DaJobBERT performed similarly when trained with sufficient in-domain data
Limitations / Research Gap	The study was limited to Danish-language data and used a dataset that was not publicly available. The model was evaluated for competence extraction and classification but was not integrated into a complete automatic job recommendation framework. Therefore, its effectiveness in other languages and in actual job-matching scenarios remains uncertain. Future research should test the model on publicly available English datasets and integrate it into a complete job recommendation system.
Relevance to AYUDA	The study is relevant to AYUDA because its BERT-based model supports automatic competence extraction and classification from job postings and candidate profiles. This can guide AYUDA's resume processing, skill normalization, and candidate-to-job matching features. However, the approach must be adapted and validated using English or Filipino recruitment data because the original model was developed using Danish-language data from Jobindex.
Implement Decision	**Adapt.** AYUDA may adapt the study's BERT-based competence extraction and classification approach for resume processing and candidate-to-job matching. The model must be retrained or validated using English or Filipino recruitment data because the original study used Danish-language data and a Jobindex dataset. Implementation should depend on data availability, technical feasibility, and system testing.
Synthesis	Li and Lioma (2023) demonstrated that a BERT-based model can jointly extract and classify competences from job postings and candidate information using ESCO categories. The model achieved strong classification performance and processed job-related text efficiently. However, it was limited to Danish-language data and was not tested in a complete job recommendation system. For AYUDA, the study provides a basis for developing NLP-assisted resume processing, skill normalization, and candidate-to-job matching. The approach should be adapted and validated using English or Philippine recruitment data before implementation. Therefore, the study supports AYUDA's proposed system by showing how standardized competence extraction can make candidate matching more organized and efficient.

RRL 10 — RINX: A System for Information and Knowledge Extraction from Resumes

Data & Knowledge Engineering · 2023

This study describes an end-to-end résumé information extraction system designed to process varied résumé structures and organize candidate information.

ScienceDirect

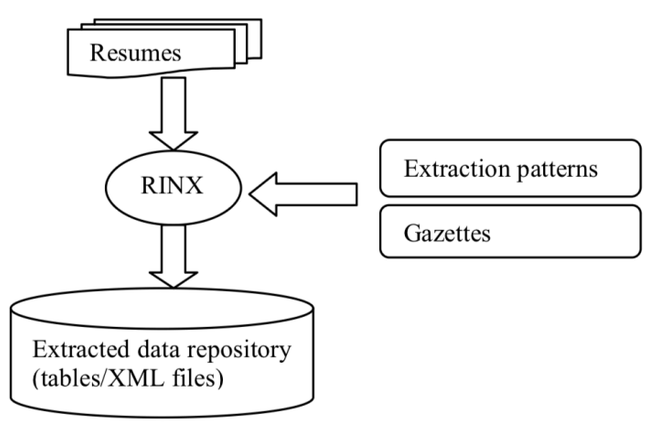
What Fiona should investigate: How extracted information is represented, whether structured knowledge is created, how different résumé layouts are handled, and how output fields could be mapped to standardized job requirements.

AYUDA relevance: Provides an additional technical foundation for creating consistent candidate information.

This is primarily an information-extraction study, not a dedicated skill-taxonomy study. Fiona should distinguish those contributions in the matrix.

Read RRL 10
https://www.researchgate.net/publication/280944384_RINX_Information_Extraction_Search_and_Insights_from_Resumes

RINX: Information Extraction, Search and Insights from Resumes

Author(s) and Year	Srivastava, R., Palshikar, G. K., & Hingmire, S. (2011). *RINX: Information extraction, search and insights from resumes*. Tata Research Development and Design Centre, TACTiCS 2011.
Research Objectives	To develop RINX, a system that extracts important information from unstructured résumés and stores it in a structured repository for candidate search, shortlisting, workforce analysis, and recruitment support.
Algorithm/Model	 **Figure 1. Information extraction functionality of RINX.** RINX uses classical information-extraction methods, including handcrafted extraction patterns, dictionaries or gazetteers, regular expressions, WordNet, and lexical, syntactic, semantic, and domain knowledge. It extracts information such as roles, service lines, technologies, education, employers, and project details. The system can also rank candidates based on a position profile and provide matching justifications.

Data Set	The verification set contained 160 free-form IT résumés. The case study processed approximately 65,000 employee résumés from a large IT organization. The résumés were mainly written in English and focused on the IT domain.
Evaluation Metrics	The study used precision, recall, and F-measure to evaluate information extraction. It also examined false positives and false negatives. For example, the extracted email information achieved 0.98 recall, 0.93 precision, and 0.95 F-measure.
Key Findings	The study found that RINX can extract and organize information from unstructured and varied résumé formats using handcrafted patterns, gazetteers, regular expressions, and linguistic and domain knowledge. It produced structured candidate information that supported résumé search, candidate shortlisting, competency analysis, workforce profiling, recruitment planning, team formation, and training decisions. However, PDF-to-text conversion errors sometimes caused important information to be missed.
Limitations / Research Gap	The approach depends heavily on manually created patterns and domain-specific dictionaries. It primarily focused on English résumés in the IT domain, so it may require adaptation for other industries, languages, and résumé formats. PDF-to-text conversion errors caused some information to be missed, and the system used only limited document-structure information.
Relevance to AYUDA	RINX supports AYUDA's centralized candidate database by extracting and organizing résumé information such as skills, roles, education, and work experience. Its standardization and search functions can help AYUDA rediscover candidates and compare candidate profiles with client job requirements.
Implement Decision	**Adapt.** AYUDA may adapt RINX's structured résumé extraction, candidate search, and skill or role standardization concepts. However, the approach must be updated for Philippine recruitment terms, different industries, modern résumé formats, OCR support, and recruiter verification.
Synthesis	Srivastava, Palshikar, and Hingmire (2011) demonstrated that RINX can extract important information from unstructured résumés and organize it into structured records using linguistic knowledge, dictionaries, handcrafted patterns, and regular expressions. The system supported candidate search, shortlisting, workforce analysis, and recruitment-related decisions. However, it mainly focused on English IT résumés and may require adaptation for other industries, languages, and résumé formats. For AYUDA, the study provides a foundation for developing centralized résumé processing, structured candidate profiles, candidate rediscovery, and job matching. The approach should be adapted to Philippine recruitment terms, modern résumé formats, OCR processing, and recruiter verification. Therefore, the study supports AYUDA's proposed system by showing how automated résumé information extraction can make candidate management and matching more organized and efficient.

Tine: Applicant-facing and bidirectional job recommendations

RRL 11 — NLP-Based Bi-Directional Recommendation System: Towards Recommending Jobs to Job Seekers and Resumes to Recruiters

Big Data and Cognitive Computing · 2022

This study proposes a bidirectional recommendation approach that recommends vacancies to job seekers and candidates to recruiters using textual information and similarity calculations.

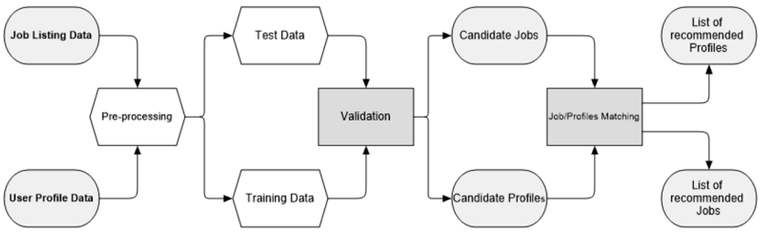
MDPI

What Tine should investigate: Its NLP pipeline, similarity method, recommendation directions, dataset, evaluation metrics, and whether explanations are provided.

AYUDA relevance: Closely resembles the three-sided system direction discussed during your professor consultation, although the paper's recommendation functions are primarily two-sided.

Read RRL 11

NLP-Based Bi-Directional Recommendation System: Towards Recommending Jobs to Job Seekers and Resumes to Recruiters

Author(s) and Year	Alsaif, S. A., Sassi Hidri, M., Ferjani, I., Eleraky, H. A., & Hidri, A. (2022). NLP-based bi-directional recommendation system: Towards recommending jobs to job seekers and resumes to recruiters. Big Data and Cognitive Computing, 6(4), Article 147. https://doi.org/10.3390/bdcc6040147
Research Objective	To develop an NLP-based bi-directional recommendation system that recommends suitable jobs to job seekers and relevant candidate resumes to recruiters. The study aims to improve recruitment matching by analyzing skills extracted from resumes and job descriptions, reducing the limitations of traditional keyword-based information retrieval.
Algorithm / Model	Figure 1 shows the flowchart of the bi-directional recommender system. The inputs of our process are job listing and user profiles data. Pre-processing and training are performed to generate candidate profiles and jobs. Finally, matching is performed to generate a list of recommended jobs for job seekers and a list of recommended profiles for employers.  **Figure 1.** Bi-directional recommender system flowchart.

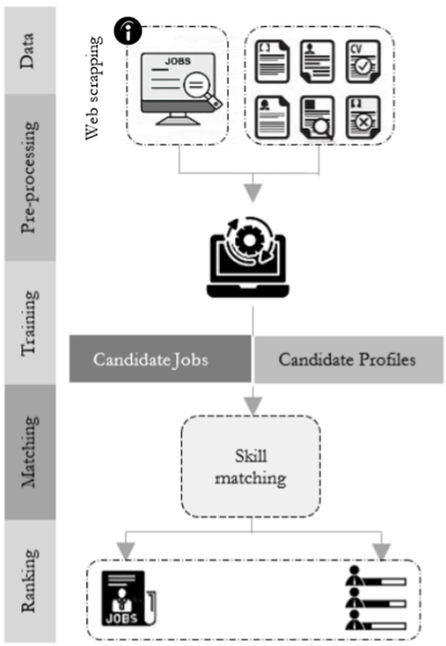


Figure 3. Overview of Bi-directional JRS.

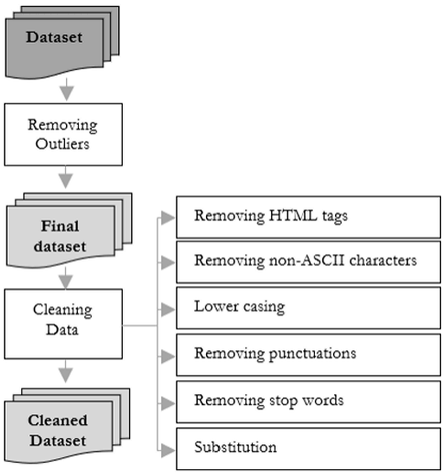


Figure 4. Overall cleaning process.

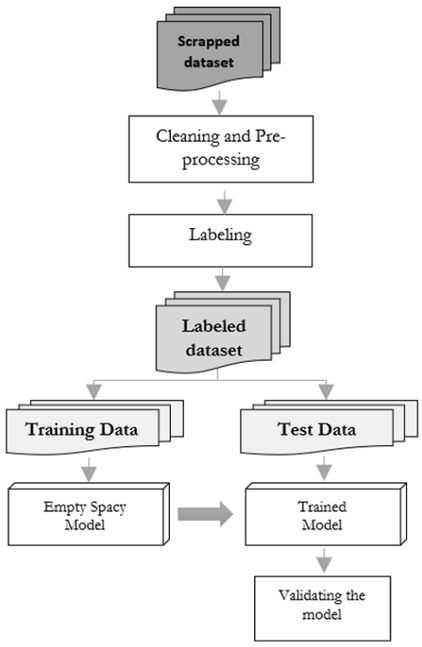
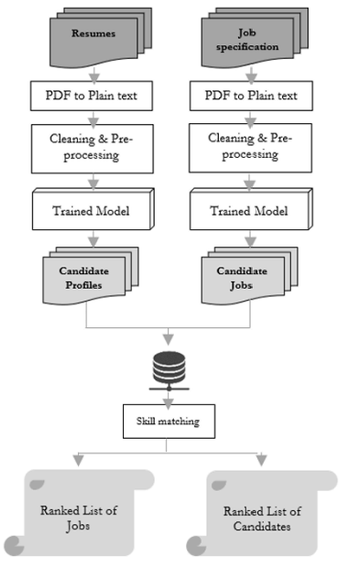


Figure 5. NLP-based training diagram.

	 **Figure 6.** Matching jobs and candidate profiles using the trained model. The researchers implemented a Content-Based Filtering (CBF) approach using Natural Language Processing (NLP). The system uses spaCy Named Entity Recognition (NER) to extract information such as skills, education, names, and locations from resumes and job descriptions. It incorporates Word2Vec and cosine similarity for measuring relationships between extracted terms and generating ranked recommendations in both directions. The workflow includes web scraping, text preprocessing, NER training, skill extraction, and similarity-based matching.
Dataset	The study used two data sources: resume profiles and job postings. 138 annotated resumes from the DataTurks GitHub dataset were used for NER training, while 25 resumes from Hire IT People were used for testing. Approximately 250 job descriptions were scraped from Indeed Saudi Arabia, covering Jeddah, Riyadh, and Dammam. The job postings represented five IT roles: PHP Developer, Data Scientist, Data Engineer, Java Developer, and Back-end Developer.
Evaluation Metrics	The researchers used Accuracy, Precision, Recall, and F1-score to evaluate entity recognition and recommendation performance. A confusion matrix was also used to assess classification errors across the five job categories. Cosine similarity was used to calculate matching scores between job profiles and candidate resumes.
Key Findings	The proposed system achieved 80% overall accuracy in classifying 25 resumes across five job categories, correctly identifying 20 resumes. The macro-average Precision and Recall were both 0.80, while the F1-score was 0.79. For entity recognition, the reported accuracy values were 99.885% for Name, 99.77% for Location, 99.08% for Skills, and 100% for Education. The system successfully generated ranked job recommendations for candidates and ranked candidate recommendations for recruiters. However, Data Engineer classification performed less effectively, with an F1-score of 0.44, partly due to overlapping skills across related IT roles.
Limitations / Research Gap	The study used a relatively small evaluation dataset of 25 resumes and focused on five IT-related positions in Saudi Arabia, limiting generalizability across industries and recruitment environments. The model also experienced difficulty distinguishing roles with overlapping skills, particularly Data Scientist and Data Engineer. The researchers identified the need to incorporate the recency of work experience, improve skill extraction using skill ontologies, and extend geographic matching to neighboring locations. Additionally, the study did not establish performance through real-world recruiter deployment or evaluate long-term recruitment outcomes.
Relevance to AYUDA	The study supports AYUDA's proposed resume parsing, skill extraction, candidate-to-job matching, and candidate rediscovery features. Its NLP-based approach demonstrates how skills and qualifications can be extracted from resumes and compared with job requirements to generate ranked

	recommendations. The recruiter-facing recommendation component is particularly relevant to identifying suitable candidates from AYUDA's existing database when new job requests become available.
Implement Decision	Adapt and Benchmark. AYUDA can adapt the study's NLP-based resume parsing, Named Entity Recognition, and skill-based matching approach. The extracted qualifications can support structured candidate profiles and recommendation explanations. However, the proposed similarity-based matching method should be benchmarked against alternative semantic matching approaches using AYUDA's available recruitment data, particularly for candidates with incomplete resumes or overlapping qualifications.
Synthesis	Alsaif et al. (2022) developed an NLP-based bi-directional recommendation system that uses Content-Based Filtering, Named Entity Recognition, Word2Vec, and cosine similarity to match candidate resumes with job postings. The system extracts relevant information, particularly skills and qualifications, to generate job recommendations for job seekers and candidate recommendations for recruiters. The proposed approach achieved 80% classification accuracy across five IT job categories and demonstrated high accuracy in extracting named entities from resumes. However, the study revealed difficulties in distinguishing job roles with overlapping skills and was evaluated using a relatively small dataset. The study provides a methodological foundation for AYUDA's proposed resume parsing, candidate profiling, and candidate-to-job matching features. Its approach demonstrates how NLP can transform unstructured recruitment documents into structured information that supports candidate recommendations and rediscovery. However, the limitations involving dataset size, overlapping qualifications, and restricted job categories indicate the need for further evaluation. Therefore, AYUDA can adapt the study's NLP-based information extraction and skill-matching techniques while benchmarking more flexible semantic matching methods to improve recommendation reliability across different job positions.

RRL 12 — Modeling Two-Way Selection Preference for Person-Job Fit

Yang et al. · ACM RecSys · 2022

This study investigates person–job fit by modeling selection preferences from both sides of the recruitment relationship, rather than considering only one party's suitability criteria.

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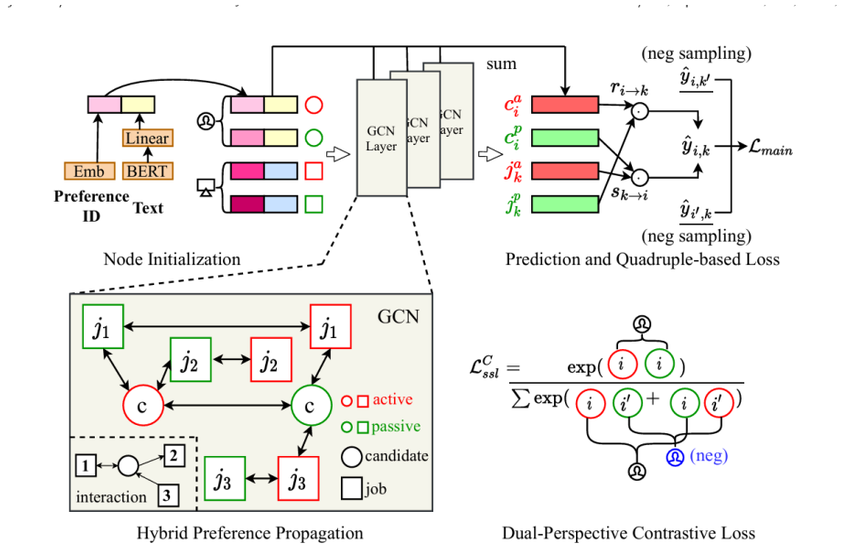
What Tine should investigate: How two-way preferences are represented, the model architecture, training data, evaluation baselines, and how reciprocal preferences differ from simply reversing the direction of recommendations.

AYUDA relevance: Helps investigate recommendations that consider both the applicant's preferences and the employer's requirements.

Read RRL 12

Modeling Two-Way Selection Preference for Person-Job Fit

Author(s) and Year	Yang, C., Hou, Y., Song, Y., Zhang, T., Wen, J.-R., & Zhao, W. X. (2022). Modeling two-way selection preference for person-job fit. In Proceedings of the 16th ACM Conference on Recommender Systems (RecSys '22, pp. 102–112). Association for Computing Machinery. https://doi.org/10.1145/3523227.3546752
Research Objective	To improve Person-Job Fit by modeling the two-way selection preferences of

	job seekers and employers, rather than relying only on resume–job similarity or one-sided recommendations. The study addresses the problem of unsuccessful matches when only one party is interested in the other.
Algorithm / Model	 The researchers proposed the Dual-Perspective Graph Neural Network (DPGNN), which combines BERT-based text representations with historical candidate–job interactions. The model constructs a dual-perspective interaction graph that represents each candidate and job through active and passive preference nodes. It uses Graph Convolutional Networks (GCN) with hybrid preference propagation, a quadruple-based ranking loss, and dual-perspective contrastive learning to predict mutual matching preferences.
Dataset	The study used three real-world recruitment datasets from the online recruitment platform BOSS Zhipin, covering 106 days of interaction logs. The Tech dataset contained 56,634 candidates, 48,090 jobs, and 925,193 successful matches. The Sales dataset contained 15,854 candidates, 12,772 jobs, and 145,066 matches. The Design dataset contained 12,290 candidates, 9,143 jobs, and 166,270 matches. Each dataset included candidate resumes, job descriptions, directed interaction records, and successful matching records.
Evaluation Metrics	The researchers used Recall@5, Precision@5, Normalized Discounted Cumulative Gain (NDCG@5), and Mean Reciprocal Rank (MRR@5) to evaluate recommendation performance in both directions: recommending jobs to candidates and recommending candidates to employers. They also conducted ablation experiments, data sparsity analysis, and hyperparameter sensitivity testing.
Key Findings	DPGNN outperformed most evaluated baseline models across the three datasets, achieving average relative improvements of 7.12% for Tech, 4.81% for Sales, and 7.73% for Design over the strongest baselines, as reported by the authors. For the Tech dataset, it achieved Recall@5 of 0.2941 for candidate recommendations and 0.3430 for job recommendations, with NDCG@5 scores of 0.3119 and 0.4410, respectively. Ablation experiments demonstrated that the dual-perspective interaction graph and quadruple-based ranking loss contributed substantially to performance. The model also performed competitively for users with sparse interaction histories.
Limitations / Research Gap	The model depends on historical candidate–job interactions, which may limit its applicability to new candidates or jobs with little behavioral data. Although the study evaluated interaction sparsity, it did not fully establish performance for completely new users without interaction histories. The researchers also identified the need to incorporate additional recruitment factors, including timestamps, salary expectations, working locations, and sequential behavior histories. Furthermore, the study did not directly evaluate fairness between candidates and employers or validate the system through long-term recruiter deployment.
Relevance to AYUDA	The study supports AYUDA's proposed candidate-to-job matching and candidate rediscovery features by demonstrating the importance of

	considering both candidate qualifications and recruitment preferences. Its use of BERT-based text representations provides a basis for semantic matching, while its dual-perspective approach highlights the potential value of considering candidate interest, employer requirements, and historical recruitment interactions. This is particularly relevant to reducing unsuitable recommendations when a candidate appears qualified but may not be interested or available for a position.
Implement Decision	Adapt and Benchmark. AYUDA can adapt the study's principle of two-way recruitment matching by considering both employer requirements and candidate preferences, such as preferred roles, locations, and availability. However, implementing the complete DPGNN architecture is not recommended without first assessing whether AYUDA has sufficient historical interaction data. The study's approach can serve as a benchmark or future enhancement, while simpler semantic matching and preference-based filtering methods may be more practical for initial implementation.
Synthesis	Yang et al. (2022) proposed a Dual-Perspective Graph Neural Network (DPGNN) to improve Person-Job Fit by modeling the selection preferences of both job seekers and employers. The model integrates BERT-based textual representations with historical recruitment interactions through a dual-perspective interaction graph, Graph Convolutional Networks, quadruple-based ranking loss, and contrastive learning. Experimental results across three real-world recruitment datasets demonstrated improvements over the evaluated baseline models, with average relative performance gains of 7.12% for Tech, 4.81% for Sales, and 7.73% for Design. The findings emphasize that effective recruitment matching should consider mutual preferences rather than relying solely on textual similarity or one-sided recommendations. The study provides a conceptual foundation for AYUDA's proposed candidate-to-job matching and candidate rediscovery features by highlighting the importance of combining candidate qualifications with employer requirements and candidate preferences. However, the model's reliance on historical interaction data and its relatively complex graph-based architecture may limit direct implementation in recruitment environments with insufficient behavioral records. Therefore, AYUDA can adapt the principle of two-way preference matching while benchmarking simpler semantic matching and preference-based filtering approaches before considering more advanced graph-based recommendation models.